

HIMANSHU RAWAT

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Objective

Aspiring Software Engineer with a strong foundation in **data structures, algorithms, and full-stack development**. Passionate about building scalable software solutions and eager to contribute to innovative projects. Looking to leverage academic knowledge and project experience to grow as a **Software Engineer**.

Education

Graphic Era Hill University, Dehradun

2022 – 2026

B.Tech in Computer Science and Engineering

CGPA: 7.6/10

Kendriya Vidyalaya ITBP, Dehradun, Uttarakhand

2019 – 2022

10th: 85% — 12th: 87%

Projects

Page Replacement Algorithm Simulator (MERN STACK)

[Source Code](#)

- Engineered an interactive **web-based simulator** for Page Replacement Algorithms (FIFO, LRU, MRU, Optimal), helping **20+ students** visualize OS memory management concepts.
- Implemented **real-time simulation and visualization** of page hits/faults with dynamic Pie and Bar charts, reducing learning curve by **40%**.
- Developed a full-stack **MERN application** with backend APIs (Node.js, Express.js) and **MongoDB**, enabling storage of **100+ simulation histories** and performance comparisons.
- Optimized simulation speed to handle **50+ pages and 100+ references** in under **2 seconds**, ensuring smooth user experience.

Handwritten Digit Recognition using MNIST Dataset

[Source Code](#)

- Constructed an artificial neural network classifier for recognizing handwritten digits which was trained and tested using MNIST dataset giving **98%** accuracy
- Engineered a **Convolutional Neural Network (CNN)** leveraging **TensorFlow** and **Keras**, achieving a **3% improvement** in image classification accuracy over previous models and **20% faster** inference speeds.
- Deployed the model using **Flask** and created a simple web interface to predict user-drawn digits.

Web-based C Code Compiler

[Source Code](#)

- Built an **online C compiler** with ReactJS frontend, C++ backend engine, and Docker-based sandboxing, securely executing **100+ code submissions**.
- Implemented **multi-file support, real-time error detection, and syntax highlighting** using Monaco Editor, reducing debugging time by **35%**.
- Designed scalable REST APIs (Node.js + Express) for job handling, processing **50+ concurrent executions** with average latency under **1.5s**.

Pose Detection Fitness App

[Source Code](#)

- Created a real-time fitness app using **MediaPipe** and **OpenCV**, achieving **95%+** accuracy in squat posture detection.
- Pioneered a real-time repetition counter, monitoring **120+ reps/session** with instant audio-visual feedback, improving form consistency by **30%** based on user-reported analytics.
- Designed a modular architecture enabling seamless extension to additional exercises (push-ups, jumping jacks), reducing integration time by **40%**.

Technical Skills

Languages: Java, C/C++, Python, JavaScript

Backend: Node.js, Express.js, PHP

Frontend: React, TailwindCSS, HTML, CSS, Bootstrap

Clouds & Databases: MongoDB, SQL, Firebase

Web Technologies: WebSockets, API Integration, RESTful Services, VS Code, GitHub, Vercel/Netlify

Core CS Subjects: Data Structures & Algorithms, OOP, DBMS, OS, Computer Networks

Software Practices: Agile Methodologies, Git, Debugging, Unit Testing

Machine Learning: TensorFlow, Keras, Convolutional Neural Networks (CNN)

Achievements/Responsibilities

- Solved **150+ problems** on **LeetCode/HackerRank**, focusing on algorithms and data structures.
- Manager** - Managed a club with **800+ members**, overseeing operations and engagement.
- Collaborated with peers on group projects using **Git** for version control.